Hugo Marquerie 13/01/2026

Lem transformada fourier gaussiana

Lema 1. *Sea g dada por $\forall x \in \mathbb{R} : g(x) = e^{-\pi x^2}$, entonces $g^\wedge = g$.*

***Demostración*:** Observamos que $g \in \mathcal{L}^1(\mathbb{R})$ y $\int_{\mathbb{R}} g(x)\,\mathrm{d}x = 1$. Calculamos:

$$\begin{aligned} g^\wedge(\xi) &= \int_{\mathbb{R}} e^{-\pi x^2} e^{-2\pi i x\xi}\,\mathrm{d}x = \int_{\mathbb{R}} e^{-\pi(x^2+2ix\xi)}\,\mathrm{d}x = \int_{\mathbb{R}} e^{-\pi(x+i\xi)^2} e^{-\pi\xi^2}\,\mathrm{d}x \\ &= e^{-\pi\xi^2} \int_{\mathbb{R}} e^{-\pi(x+i\xi)^2}\,\mathrm{d}x = e^{-\pi\xi^2} \int_{\mathbb{R}} e^{-\pi y^2}\,\mathrm{d}y = e^{-\pi\xi^2} \cdot 1 = g(\xi). \end{aligned}$$

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